

Bluestock MF Analytics Platform

End-to-End Mutual Fund Intelligence for the Indian Market

Streamlit Dashboard | SQLite Star Schema | Performance Analytics

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github.com/prabhjotkaurarora27/bluestock_mf_capstone

Project Overview

40

Fund Schemes

Across 10 AMCs

64,320

NAV Records

Jan 2022 - May 2026

32,778

Transactions

Investor data

~12.9 MB

DB Size

SQLite star schema

7

Dashboard Pages

Streamlit + Plotly

Project Objectives

- Ingest & clean raw NAV, AUM, SIP, and transaction data from public AMFI sources
- Build a SQLite star-schema database with 11 tables, 107,461 rows, FK constraints and 8 indexes
- Perform EDA generating 15 charts: trends, demographics, geography, sector, correlations
- Compute CAGR, Sharpe, Alpha, Beta, Max Drawdown and Bluestock Composite Score for 40 funds
- Deliver a 7-page interactive Streamlit dashboard with Monte Carlo simulation and Markowitz frontier

Technology Stack

Layer	Technology	Purpose
Language	Python 3.9+	Core processing and analytics
Dashboard	Streamlit 1.50 + Plotly 6.7	7-page interactive web app
Database	SQLite + SQLAlchemy	Star schema, 11 tables
Data Processing	Pandas 2.3 + NumPy 2.0	ETL pipeline and cleaning
Financial Math	SciPy 1.13	Monte Carlo, Markowitz optimization
Visualisation	Plotly + Matplotlib + Seaborn	Charts and EDA
Excel Export	openpyxl 3.1	Download buttons on dashboard
Version Control	Git + GitHub	Full commit history (14 commits)

GitHub Repository

https://github.com/prabhjotkaurarora27/bluestock_mf_capstone

Table	Type	Rows	Description
dim_fund	Dim	40	Fund master - AMC, category, benchmark, manager
dim_date	Dim	1,608	Date dimension - year, quarter, month, week
fact_nav	Fact	64,320	Daily NAV per fund
fact_transactions	Fact	32,778	SIP / Lumpsum / Redemption records
fact_performance	Fact	40	1yr/3yr/5yr returns, Sharpe, Alpha, Beta
fact_aum	Fact	90	Quarterly AUM by fund house
fact_sip_inflows	Fact	48	Monthly SIP stats and account counts
fact_category_inflows	Fact	144	Category-wise net inflows
fact_folio_count	Fact	21	Monthly folio count by category
fact_portfolio_holdings	Fact	322	Sector holdings per fund
fact_benchmark_indices	Fact	8,050	Daily NIFTY 50 / NIFTY 100 close values

Total: 107,461 rows | 3 analytical views | 8 indexes | FK + CHECK constraints

Performance Analytics (Day 4)

Metric	Formula	Output File
CAGR (3yr)	$(\text{End NAV} / \text{Start NAV})^{(1/3)} - 1$	fund_scorecard.csv
Sharpe Ratio	$(R_p - R_f) / \text{StdDev} \text{ [} R_f = 6\% \text{ p.a.]}$	fund_scorecard.csv
Jensen's Alpha	$R_p - [R_f + \text{Beta} \times (R_m - R_f)]$ vs NIFTY 100	alpha_beta.csv
Beta	$\text{Cov}(R_p, R_m) / \text{Var}(R_m)$	alpha_beta.csv
Max Drawdown	$(\text{Trough} - \text{Peak}) / \text{Peak}$	fund_scorecard.csv
Bluestock Score	40% CAGR + 30% Sharpe + 20% Alpha + 10%(1-DD)	fund_scorecard.csv

Top Funds by Bluestock Composite Score

- #1 HDFC Top 100 Fund | 3yr CAGR: 18.2% | Sharpe: 1.55 | Score: 78.4
- #2 UTI Mid Cap Fund | 3yr CAGR: 21.1% | Sharpe: 1.42 | Score: 75.1
- #3 Axis Small Cap Fund | 3yr CAGR: 19.8% | Alpha: +4.1% | Score: 73.8
- #4 ABSL Small Cap Fund | 3yr CAGR: 17.6% | Beta: 0.91 | Score: 72.5
- #5 Axis Bluechip Fund | 3yr CAGR: 15.3% | Max DD: -18% | Score: 71.2

Page	Key Features
Home	Hero card, 5 KPI stats, page guide, methodology, glossary, FAQ
Industry Overview	SIP AUM dual-axis trend, AMC bar chart, folio area chart, YoY badges, CSV/Excel export
Fund Performance	Risk-return scatter, scorecard table, NAV vs NIFTY 100, fund comparison tool, drill-through
Investor Analytics	State bar chart, transaction donut, SIP by age group, T30/B30 split, state drill-through
SIP & Market Trends	SIP vs NIFTY 50 dual-axis, category heatmap, FY25 top categories, YoY growth
Monte Carlo Sim	10,000-path 5yr projection, confidence bands, histogram, P(target CAGR)
Portfolio Optim.	Markowitz frontier, 3,000 random portfolios, Max Sharpe, Min Vol, weight sliders

Run the Dashboard

```
python3 -m streamlit run streamlit_dashboard.py -> http://localhost:8501
```

Project Deliverables Summary

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Day	Deliverable	Status
Day 1	Project structure, Git init, data ingestion scripts	COMPLETE
Day 2	Data cleaning (13 CSVs) + SQLite star-schema DB loaded	COMPLETE
Day 3	Exploratory Data Analysis - 15 charts exported to reports/charts/	COMPLETE
Day 4	Performance analytics: Sharpe, Alpha, Beta, CAGR, Scorecard	COMPLETE
Day 5	7-page Streamlit dashboard + Monte Carlo + Efficient Frontier	COMPLETE

Key Outcomes

- Built a production-grade ETL pipeline processing 107,461 rows across 11 DB tables
- Delivered a professional 7-page interactive analytics platform with 20+ Plotly charts
- Implemented Monte Carlo simulation (10,000 paths) and Markowitz efficient frontier
- All deliverables committed to GitHub with full commit history (14 commits)

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